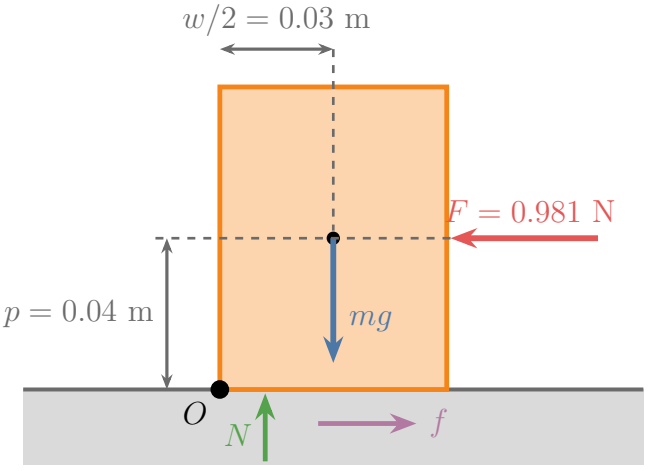
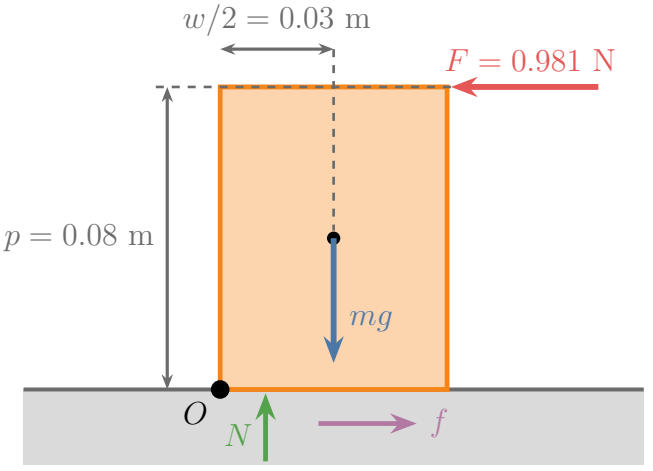


push: $0.981 \times 0.04 = 0.0392$ N m
weight: $2.4525 \times 0.03 = 0.0736$ N m
the cup slides



push: $0.981 \times 0.08 = 0.0785$ N m
weight: $2.4525 \times 0.03 = 0.0736$ N m
the cup tips